MILITARY SPECIFICATION

INDICATOR, POSITION, HORIZONTAL STABILIZER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of 28V dc, horizontal stabilizer position indicator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-T-5583	Transmitter, Position, 28-Volt DC
MIL-F-6836	Flange, Mounting, Attachable, Aircraft Instrument
MIL-I-7062	General Specification For

STANDARDS

Military

AN5786	Transmitter-Position, 28 Volts DC
AN5787	Transmitter-Position, Gearless
MIL-STD-129	Marking For Shipment And Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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MIL-STD-794	Parts and Equipment, Procedures For Packaging And Packing Of
MIL-STD-1651	Insert Arrangements For MIL-C-5015, MIL-C-22992, (Class C, J and R), And MIL-C-83723 (Series II) Electrical Connectors
MS28013	Indicator-Position; Horizontal Stabilizer
MS28055	Flange, Mounting, Attachable, Aircraft Instrument
MS33638	Case, Instrument, Flange Mounted, Aircraft

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (See 4.2 and 6.4).

3.1.1 General. The requirements specified in Specification MIL-I-7062 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2 Design and Construction. The indicator shall be designed for operation with a position transmitter in accordance with Specification MIL-T-5583.

3.2.1 Resistance. The direct current resistances of the indicators measured between the connector pins A, B, and C shall be 475 ± 100 ohms. Deviations from the averages of the resistance values between the three coils shall not exceed ±1,500 percent.

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3.2.2 Outline Dimensions. The outline dimensions of the indicator shall conform to Standard MS28013.

3.2.3 Case. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator and shall not exceed an over-all length of 2.50 inches.

3.2.3.1 Design. The design on the indicator case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the contracting service. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.2.3.2 Helium. The case shall be filled with helium of at least 98 percent purity, free of dust particles, and containing not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The absolute pressure of the helium in the case shall be approximately one atmosphere.

3.2.4 Dial. The dial shall conform to figure 1. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. The diameter of the dial shall be not less than 1.750 inches when measured across the outside ends of the graduations.

3.2.5 Cover Glass Position. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed .031 inch.

3.2.6 Pointer Overlap. The pointer length shall be such that the pointer tip will extend into the scale a distance equal to .333 to .667 of the length of the shortest graduations.

3.2.7 Electrical connector. The indicator shall be provided with an electrical connector in accordance with MIL-I-7062. The insert shall be in accordance with the 10SL-3P arrangement in MIL-STD-1651. A receptacle designed for hermetic sealing may be used but must also meet the other requirements of MIL-I-7062.

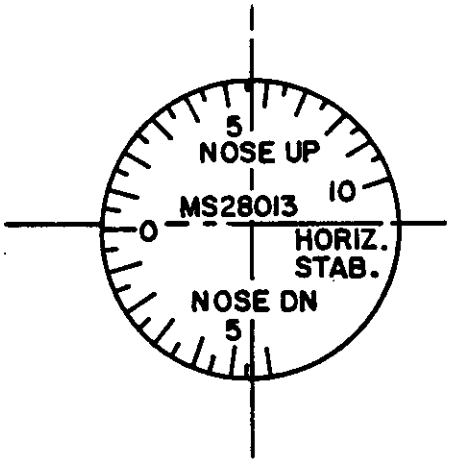
3.3 Identification of Product.

3.3.1 Nameplate. The nameplate shall be marked with the following information:

INDICATOR, POSITION HORIZONTAL STABILIZER
28 Volts DC
MS Part No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Stock No.
Manufacturer's Name or Trade-Mark
US Property

3.4 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

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MARKING	HEIGHT OR LENGTH ±0.016	WIDTH OF LINE OR GRADUATION ±.005	MATERIAL OR FINISH
NUMERALS 0, 5, AND 10 LETTERING "HORIZ STAB" LETTERING "NOSE, UP, DN" MAJOR GRADUATIONS MINOR GRADUATIONS	0.156 0.125 0.109 0.188 0.094	0 020 0.020 0.016 0.031 0.020	FLUORESCENT LUMINESCENT MATERIAL
BACKGROUND OF DIAL MS PART NO.	--- 0.063	--- 0.010	DURABLE DULL BLACK

DIMENSIONS IN INCHES

FIGURE 1. Dial and markings.

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3.3.2 Use of MS or MIL Designations. MS or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence or sales matter, until notification has been received from the Qualifying activity that the product has been approved for aeronautical use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection will be as specified in MIL-I-7062, except that where qualification is specified, first article shall apply.

b. Quality conformance inspection method of sampling and inspection shall be as specified in MIL-I-7062.

4.3 Voltage Off. With no voltage applied to the indicator, the pointer shall move off scale a minimum distance equal to the arc of 1.500 degrees of graduation.

4.3.1 Scale error. The transmitter shaft shall be rotated so that the indicator pointer is in the "0" position. The shaft shall then be rotated in 16-degree increments, and the indication shall be noted at each 2-degree graduation for full scale travel left and return. The indicator may be tapped lightly during this inspection. The error, as noted in the Scale Error inspection shall not exceed 0.250-indicated degree in the range of four (4) degrees up to four (4) degrees down, and 0.500-indicated degree over the balance of the scale.

4.3.2 Tolerances. Tolerance for subsequent inspections, in accordance with Specification MIL-I-7062, shall not exceed the following values:

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Inspections	Tolerance
Friction	0.250 degree indication
Position Error	0.250 degree indication
Low Temperature Operation	0.500 degree indication
High Temperature Operation	0.500 degree indication
Vibration Error	0.250 degree indication
Voltage Variation	0.250 degree indication

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test condition specified in MIL-I-7062.

4.5 Inspection of packaging. The inspection of the preservation and interior package marking shall be in accordance with group A and B quality conformance requirements Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provision of applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial as specified. (see 6.3)

5.1.1 Level A

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Methods IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.3).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless

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otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords protection from domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.3), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The position indicator covered by this specification is intended to indicate the position of the horizontal stabilizer in degrees of change from the neutral or zero position, and the reaction on the aircraft due to the change, such as tending to position the nose up or down. This reaction of the stabilizer setting on the aircraft is the primary indication.

6.2 Horizontal stabilizer. The horizontal stabilizer position indicator may be installed in instrument panels drilled for the instrument cases in accordance with MS33638 by use of the MS28055 instrument mounting bezel, conforming to Specification MIL-F-6836.

6.3 Ordering data. Requisitions, contracts, and order should specify the part numbers of the indicators which are to be procured, and whether overseas packing is desired.

6.4 First article. When a first article inspection is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, text and approval of the first article.

6.5 Subject Term (key word) listing.

Horizontal Stabilizer
Indicator
Position
28V DC

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6.6 Changes from previous issues. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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